

UiPath Tutorial: Agentic Automation for Appointment Emails

Introduction

We are using UiPath Studio Web for this tutorial.

Students should first go to the UiPath Studio Web URL, log in, and create an account if they do not already have one.

Objective

We are building an AI-powered automation agent that can:

1. Read customer emails from a Google Sheet
2. Understand the request (e.g. confirm, cancel, reschedule)
3. Check appointment records
4. Send an automated reply

In simple terms, we are replacing manual email handling with an automated assistant.

Part 1: Understand the Data

Step 1.1: Create your own Google Sheet named *customer_emails*.

- Use the file in the link as the source data, and copy it into your sheet.
<https://bit.ly/3Rg7LoH>
- This sheet contains the email address, subject, and email body. It will be used to simulate incoming customer emails.

Step 1.2: Create your own Google Sheet named *appointments*.

- Use the file in the link below as the source data, and copy it into your sheet.
<https://bit.ly/4unp0CO>
- This sheet contains the customer name and appointment details. It will act as the appointment database.

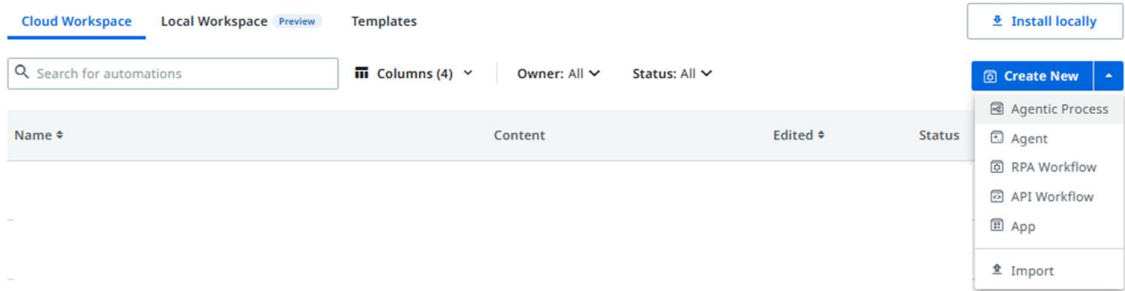
These two spreadsheets are the main data sources used in the workflow.

Alternatively, you may copy the spreadsheet from Google drive or NTULearn site.

1. Create UiPath Workflow

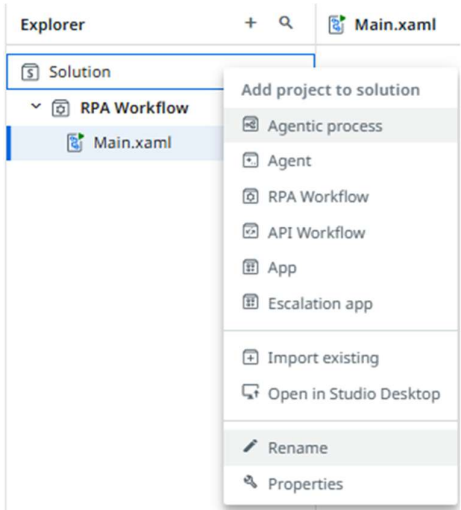
1.1. Open UiPath Studio Web

Go to UiPath Studio Web and click **Create New** project, choose **RPA Workflow**



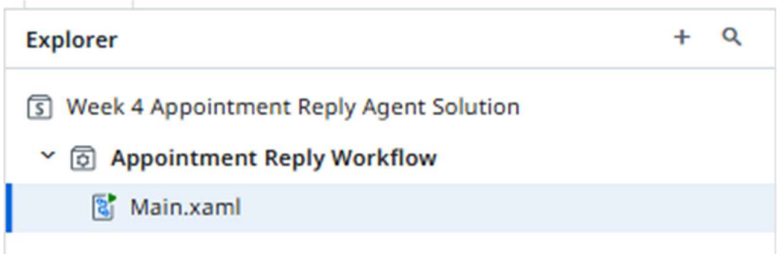
1.2. Rename the solutions and workflow by right clicking on their names

Original Name	New Name
Solution	Appointment Reply Agent Solution
RPA Workflow	Appointment Reply Workflow



1.3. Rename the solutions and workflow by right clicking on their names

This is how they should look like after you rename them



2. Read Email Data from Google Sheets

2.1. Add “For Each Row in Spreadsheet (Google Workspace)” activity

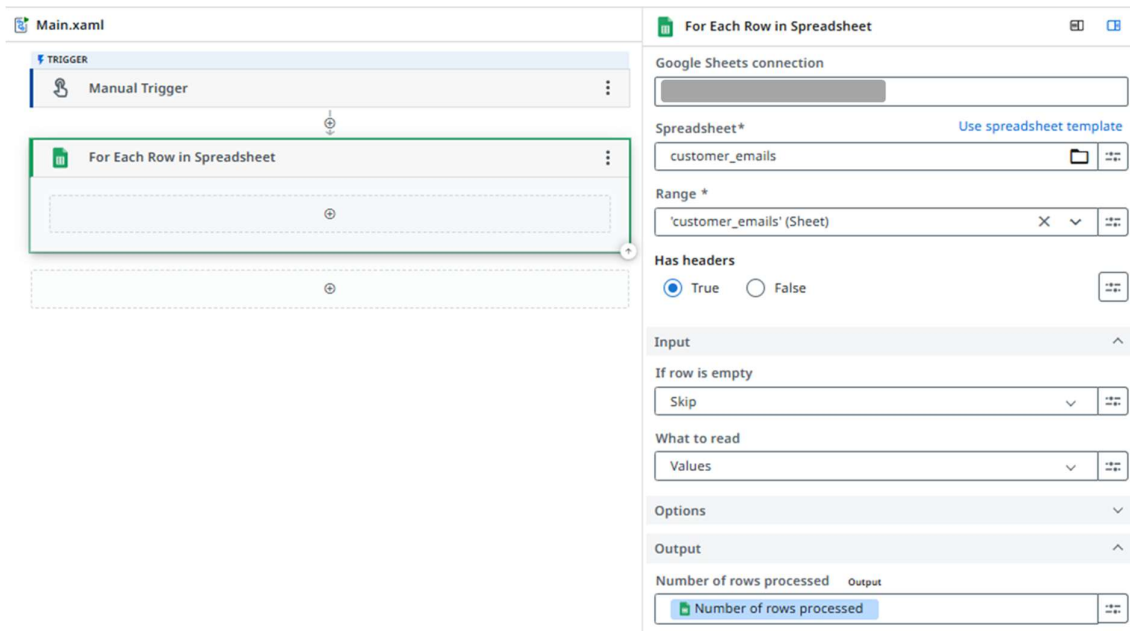
The screenshot shows the Zapier interface for configuring a workflow. At the top, a search bar contains the text "For each row in spreadsheet". Below the search bar, a card for the "For Each Row in Spreadsheet" activity is displayed, with the description "Reads and saves the information from a spreadsheet's range for subsequent activ..." and the Google Workspace logo.

The main workspace shows a workflow named "Main.xml" with a "Manual Trigger" activity followed by the "For Each Row in Spreadsheet" activity. The configuration panel for the "For Each Row in Spreadsheet" activity is open on the right, showing the following settings:

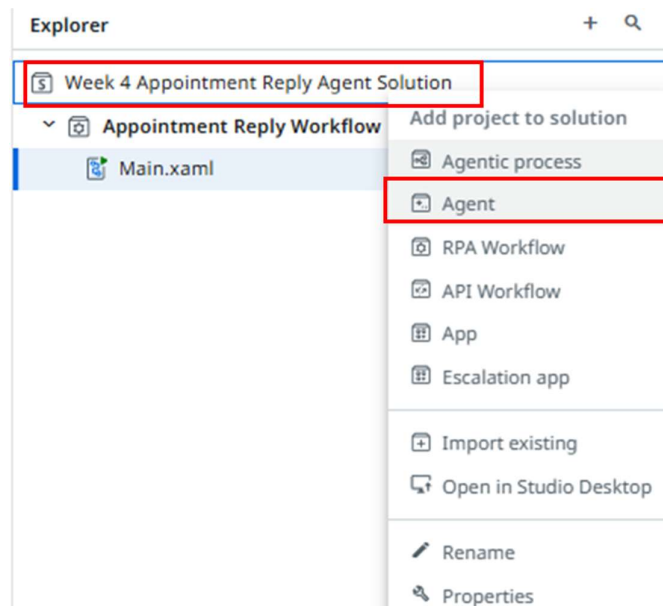
- Google Sheets connection:** A dropdown menu with a placeholder text.
- Spreadsheet*:** A dropdown menu with a placeholder text and a link to "Use spreadsheet template".
- Range*:** A dropdown menu with a placeholder text "Choose or type a value".
- Has headers:** Radio buttons for "True" (selected) and "False".
- Input:** A section with a dropdown menu for "If row is empty" set to "Skip".
- What to read:** A dropdown menu set to "Values".
- Options:** A section with a dropdown menu.
- Output:** A section with a dropdown menu.
- Number of rows processed:** A dropdown menu set to "Number of rows processed".

2.2. Fill in the details

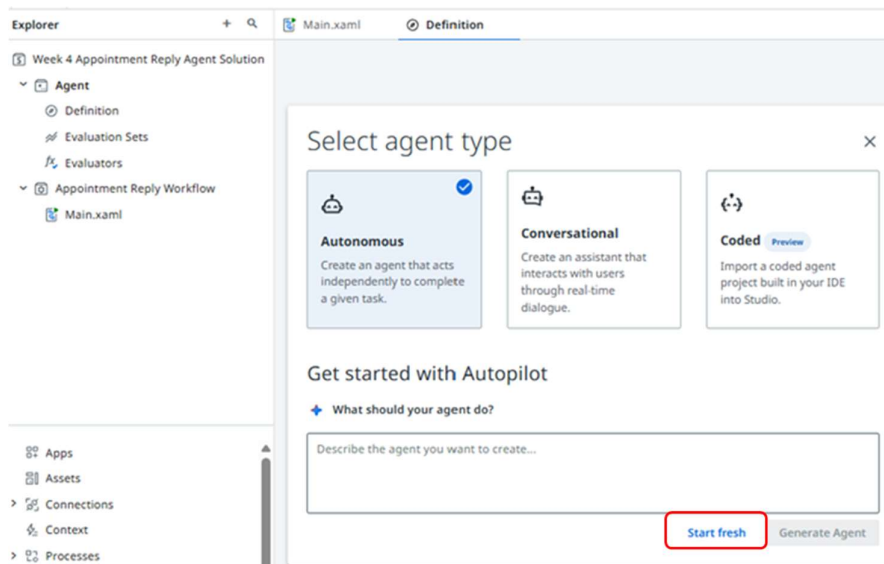
- Google Sheets connection: Add a connection to your own Google Sheet
- Spreadsheet: Browse through your Google Drive and choose “customer_emails” Google Sheet file
- Range: Choose your sheets in the Google sheet file (it should either be “customer_emails” or “sheet1” depending on how you name it)
- Has headers: the default is True, leave it as it is



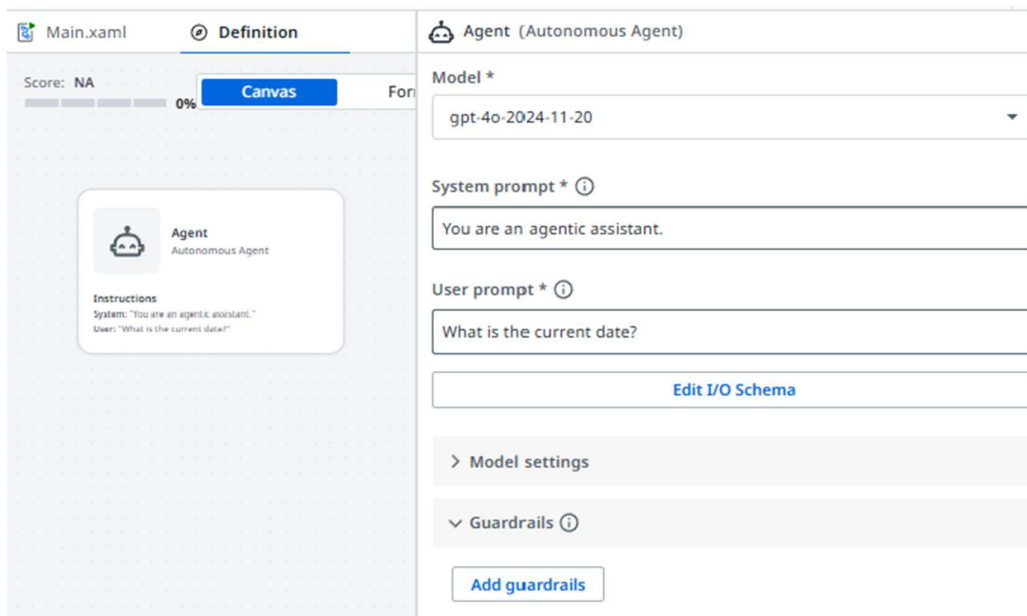
3. Right Click “Appointment Scheduler Agent Solution” and choose “Add project to solution” → “Agent”



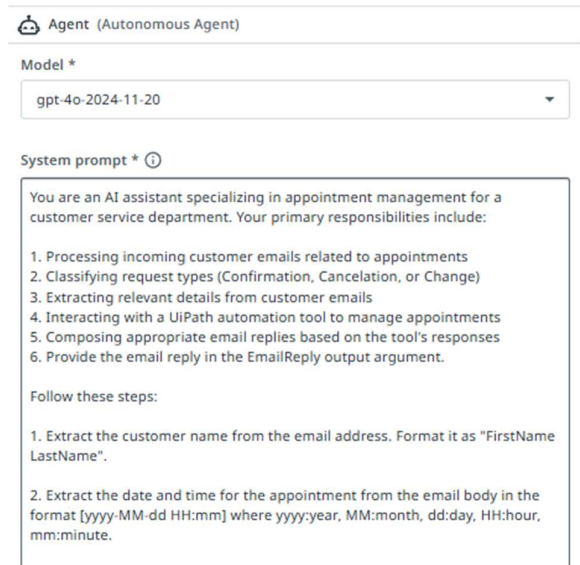
3.1. Click “Start fresh”



You should see a screen similar to the following:



3.2. Copy content from text file in the linked below into the “System Prompt”
(Source: <https://bit.ly/4bS0MdN>)



Agent (Autonomous Agent)

Model *

gpt-4o-2024-11-20

System prompt * ⓘ

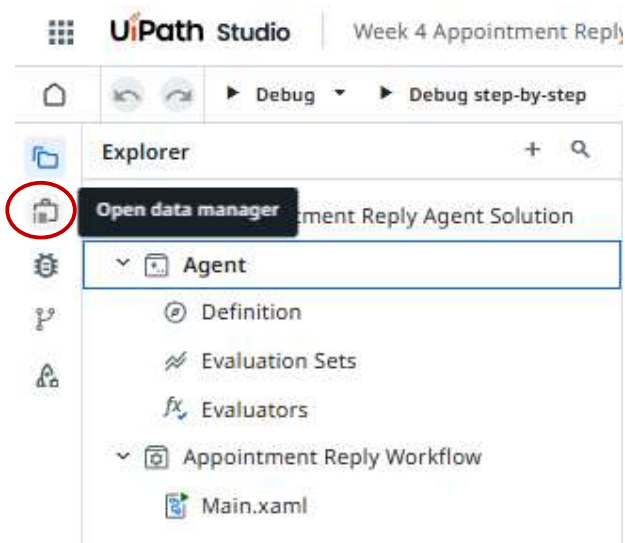
You are an AI assistant specializing in appointment management for a customer service department. Your primary responsibilities include:

1. Processing incoming customer emails related to appointments
2. Classifying request types (Confirmation, Cancellation, or Change)
3. Extracting relevant details from customer emails
4. Interacting with a UiPath automation tool to manage appointments
5. Composing appropriate email replies based on the tool's responses
6. Provide the email reply in the EmailReply output argument.

Follow these steps:

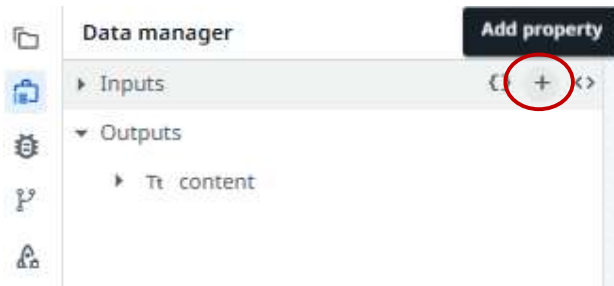
1. Extract the customer name from the email address. Format it as "FirstName LastName".
2. Extract the date and time for the appointment from the email body in the format [yyyy-MM-dd HH:mm] where yyyy:year, MM:month, dd:day, HH:hour, mm:minute.

3.3. Go to the data manager (mouse over icon shown below and click it)

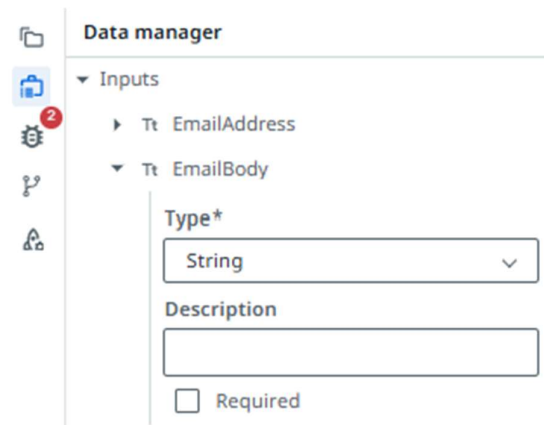


/.

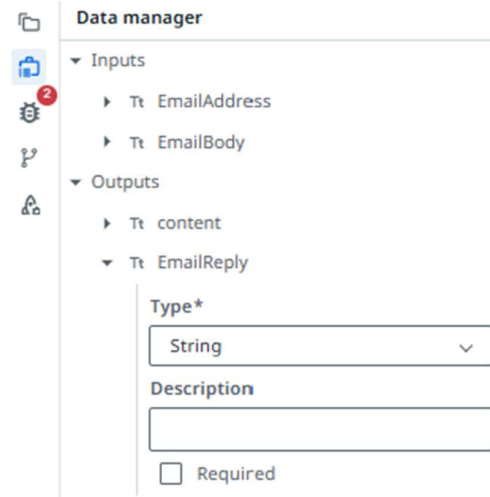
3.4. Go to **Inputs** and click on **+** to create two new variables



Name the two variables **EmailAddress** and **EmailBody**, keep the data type as **String**



3.5. Go to **Outputs and click on + to create one new variable **EmailReply****



- 3.6.** Copy content from text file in the linked below into the “User Prompt” section
(Source: <https://bit.ly/41hKxR1>)

User prompt * ⓘ

Process the following customer email request:

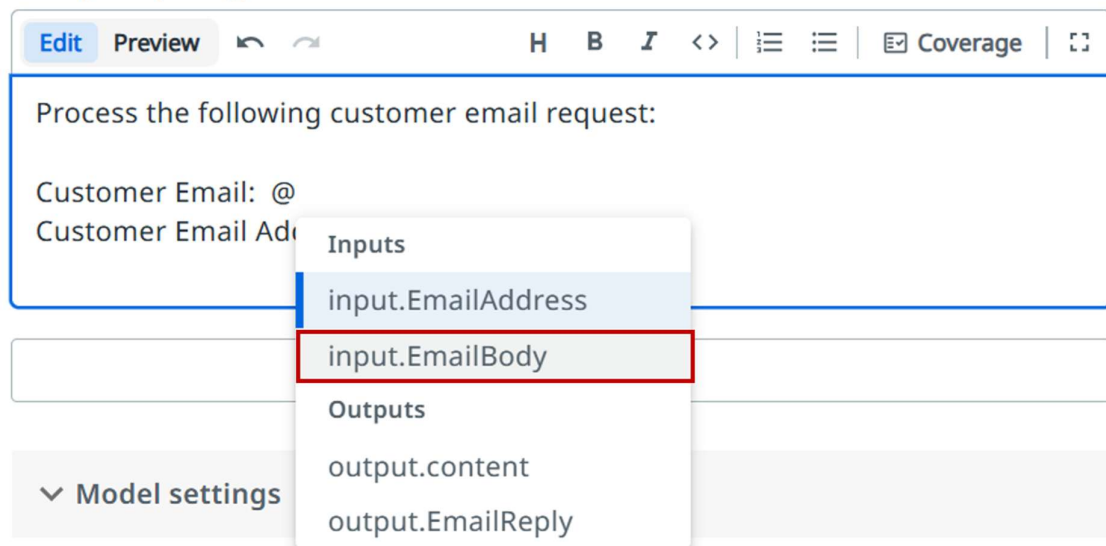
Customer Email: {{EmailBody}}

Customer Email Address: {{EmailAddress}}

- 3.7.** Remove {{EmailBody}} and {{EmailAddress}}

- 3.8.** After “Customer Email:”, type “@”. A dropdown menu will appear, choose
“input.EmailBody”

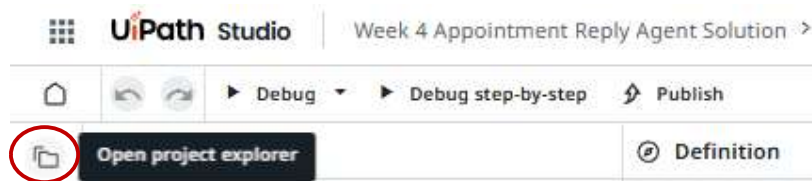
User prompt * ⓘ



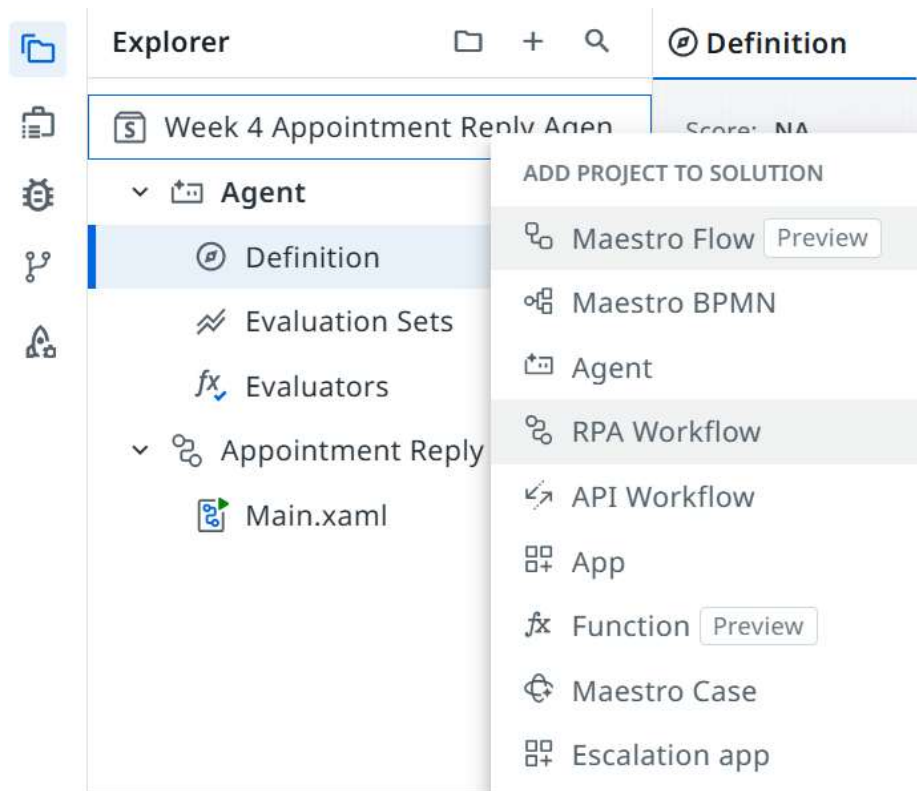
- 3.9.** Do the same for “Customer Email Address:”, choose “input.EmailAddress”

4. Add RPA Workflow - Appointment Check

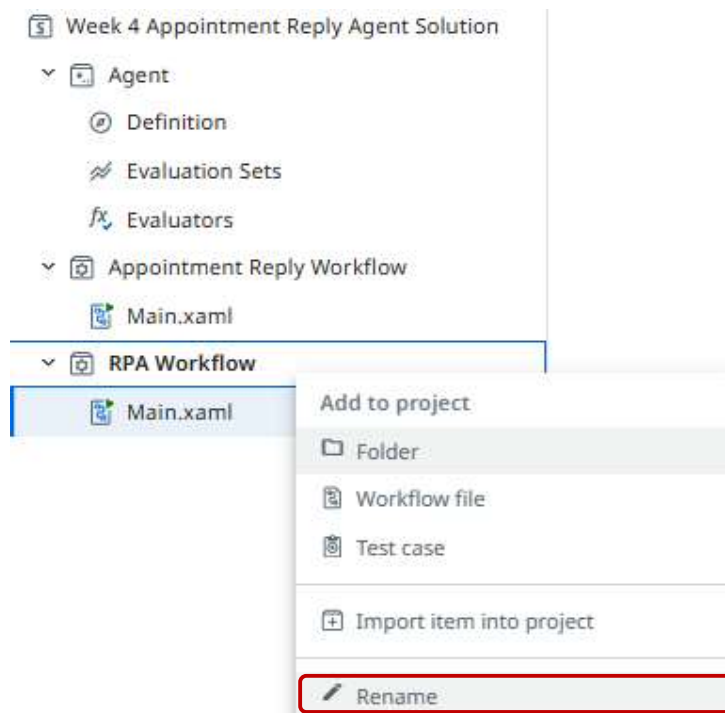
4.1. Click open project explorer icon



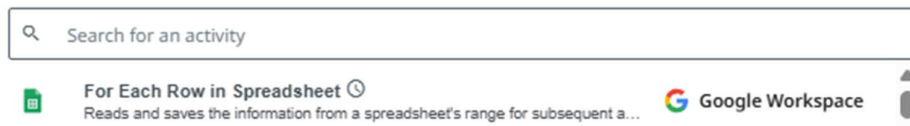
- 4.2.** Right Click “Appointment Scheduler Agent Solution” and choose “Add RPA workflow”



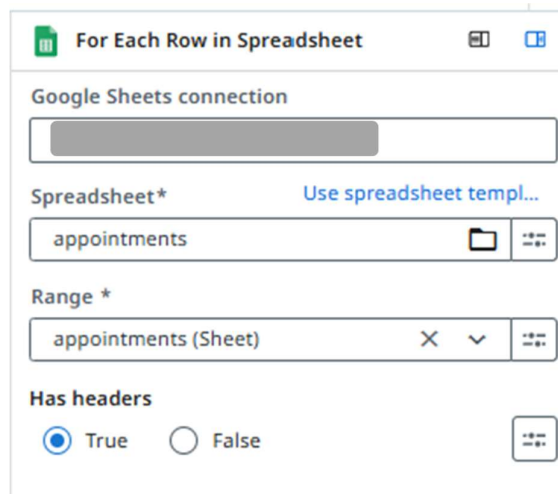
Right click **RPA Workflow** and rename it to **Appointment Check Workflow**



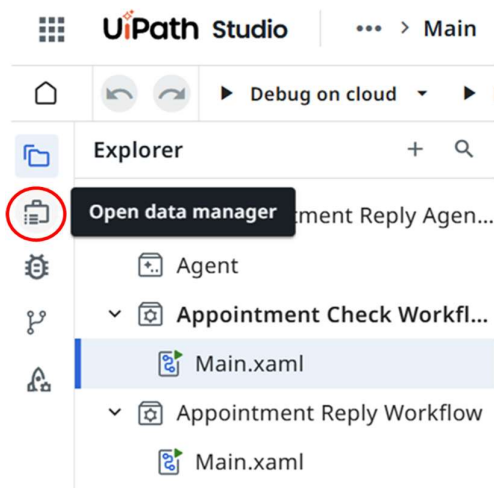
4.3. Add Activity “For each row in spreadsheet (Google Workspace)”



In the right panel, choose **appointments** for spreadsheet field and **appointments (Sheet)** for Range field

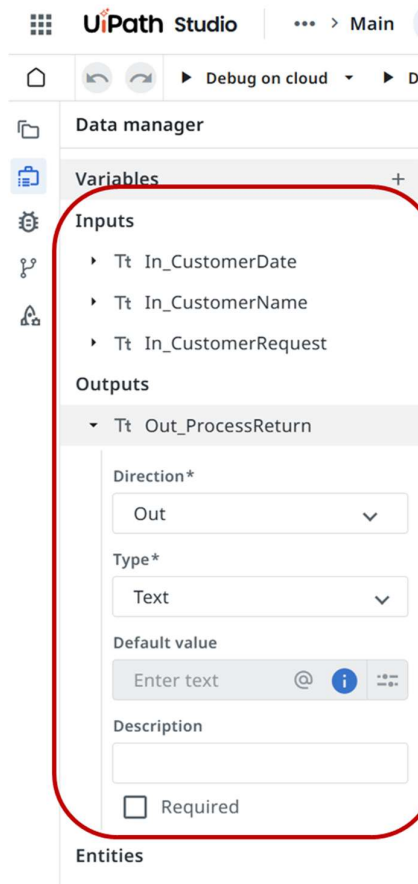


4.4. Click on open data manager to create new arguments



Create 3 input arguments and 1 output argument:

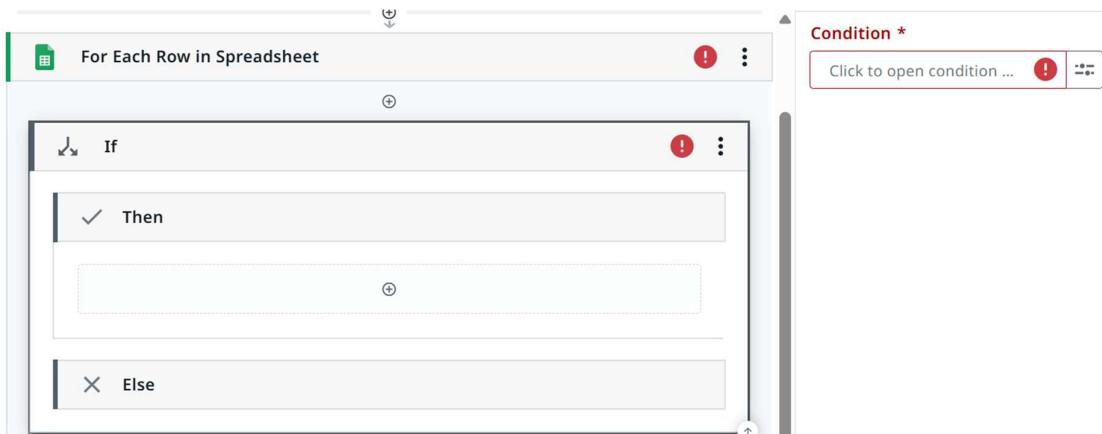
- Input: In_CustomerDate (Type: Text)
- Input: In_CustomerName (Type: Text)
- Input: In_CustomerRequest (Type: Text)
- Output: Out_ProcessReturn (Type: Text)



4.5. Add an “If” activity



Click “Click to open the condition builder” (right)



Condition: CurrentRow.Name (click see more → variable → Name) equals
In_CustomerName (click see more → variable → In_CustomerName)

Condition builder

✓ All (AND)

Any (OR)

of the conditions are met:

{x} Current row Name X

⋮

equals

⌵

{x} In_CustomerName X

⋮

🗑

Add

⌵

Cancel

Save

Click “Save”

4.6. Add a “Switch” statement inside the Then block

TRIGGER

Manual Trigger

⋮

⌵

For Each Row in Spreadsheet

⋮

⌵

If

⋮

✓ Then

⌵

Switch

⋮

⊕ Add child

Default

⊕

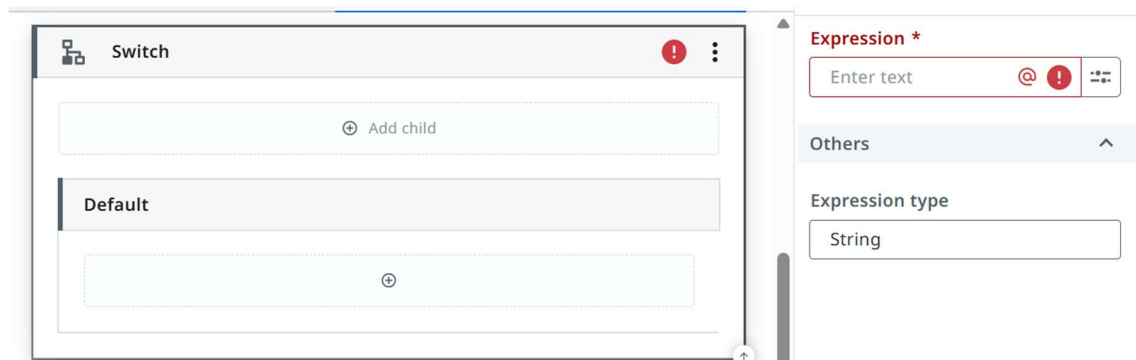
⌵

✕ Else

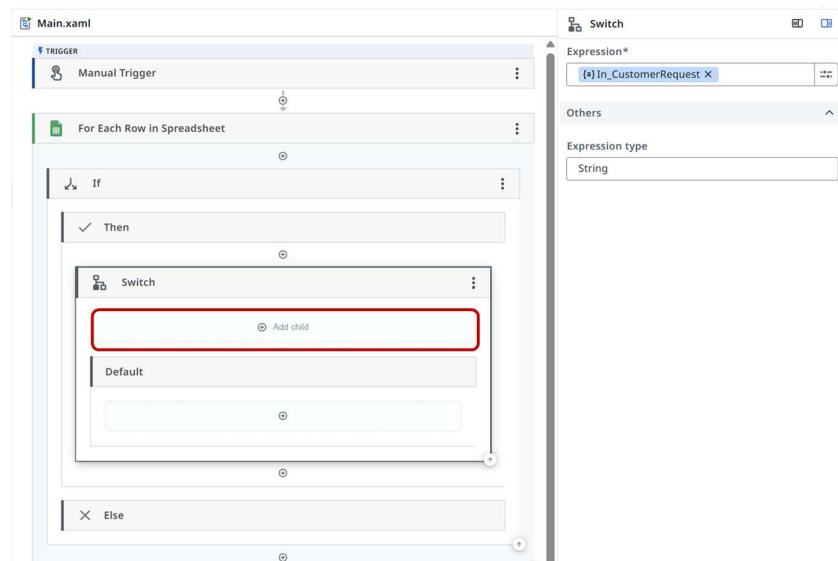
⌵

⌵

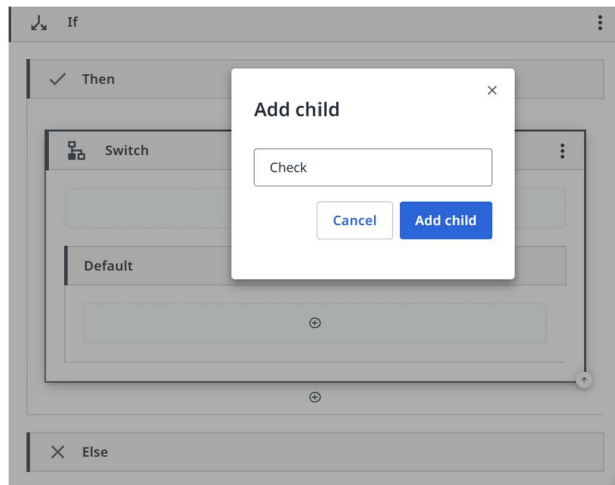
In the expression field, click on see more → variable → select **In_CustomerRequest**



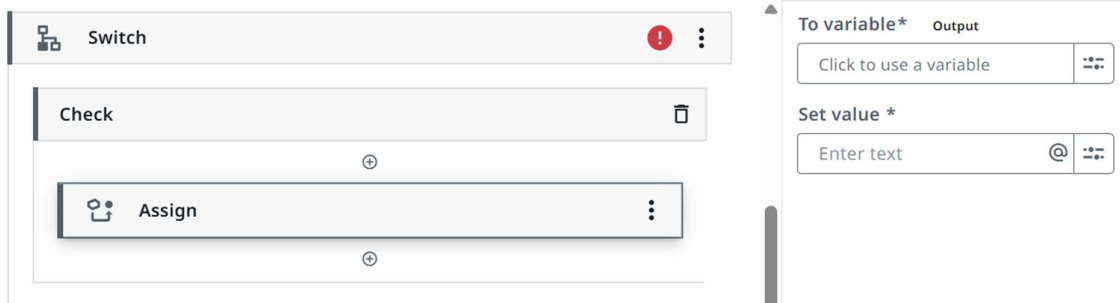
In the switch statement block, click on **Add child**



Name the first child **Check**, click on **Add child**

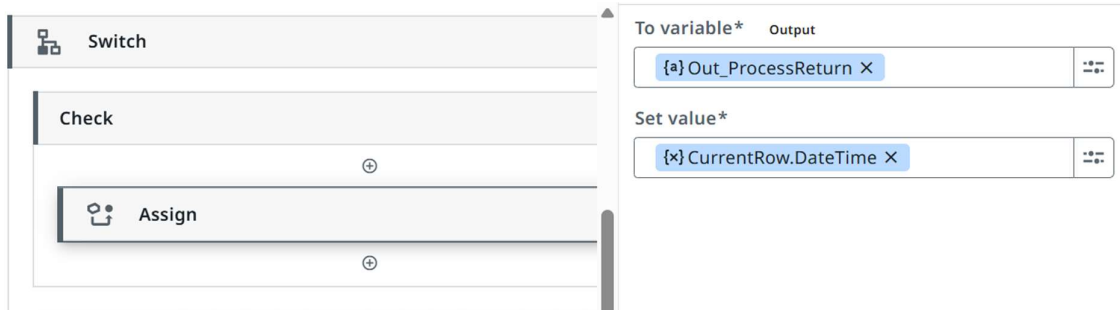


Inside **Check**, add a new activity **Assign**



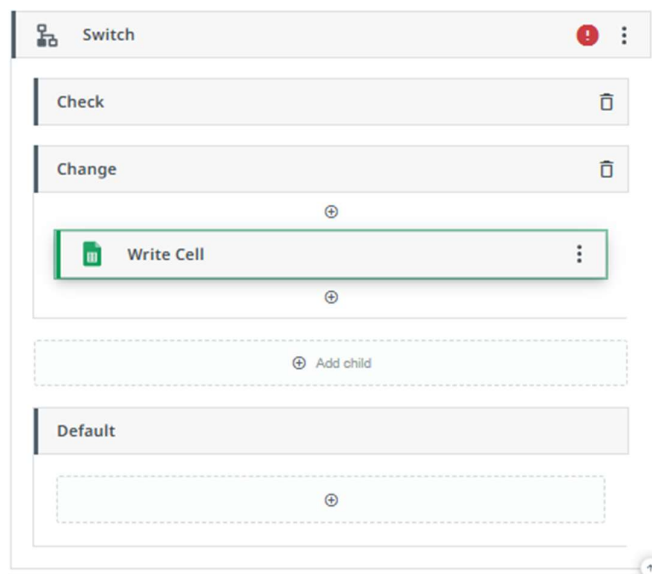
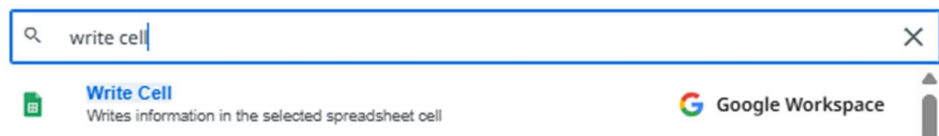
In the **To variable** field, click see more → Change variable → select **Out_ProcessReturn**

In the **Set value** field, click see more → variable → select **Date time**

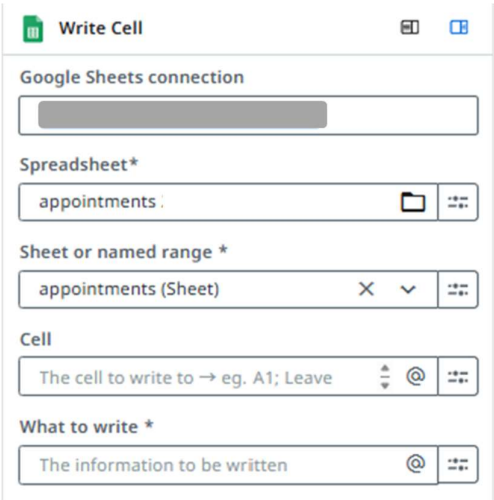


Add a second child, name it **Change**

Add a **Write Cell** activity



Fill in the cell details for **Spreadsheet** (appointments) and **Sheet or named range** (appointments (Sheet))



Write Cell

Google Sheets connection

Spreadsheet*

appointments

Sheet or named range *

appointments (Sheet)

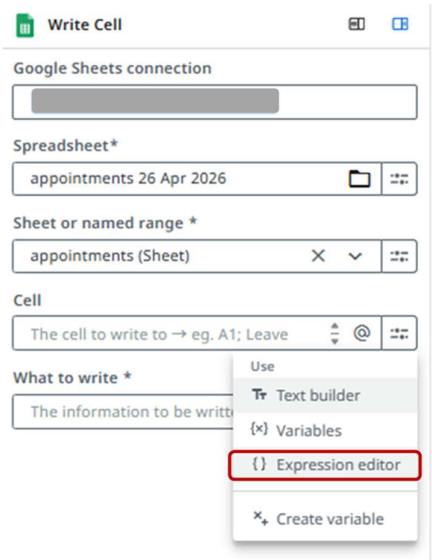
Cell

The cell to write to -> eg. A1; Leave

What to write *

The information to be written

For the **Cell** field, right click **see more** → select **Expression editor** and type “B” & CurrentRowIndex+2. Click **Save**.



Write Cell

Google Sheets connection

Spreadsheet*

appointments 26 Apr 2026

Sheet or named range *

appointments (Sheet)

Cell

The cell to write to -> eg. A1; Leave

What to write *

The information to be written

- Use
- Text builder
- Variables
- Expression editor**
- Create variable



VB Expression editor Write Cell > Cell (Text String)

Fix {x} Insert variable

```
1 "B" & CurrentRowIndex+2
```

Describe your expression here. Use Ctrl+Space to insert variables.

For the **What to write** field, click **See more** → **variable** → **in_CustomerDate**

Write Cell

Google Sheets connection

Spreadsheet*

appointments 26 Apr 2026

Sheet or named range *

appointments (Sheet)

Cell

"B"

&

CurrentRowIndex+2

X

What to write *

The information to be written

Search

Assign Variable Value

{}

To variable →

Out_ProcessReturn

> For Each Row in Spreadsheet

> Manual Trigger

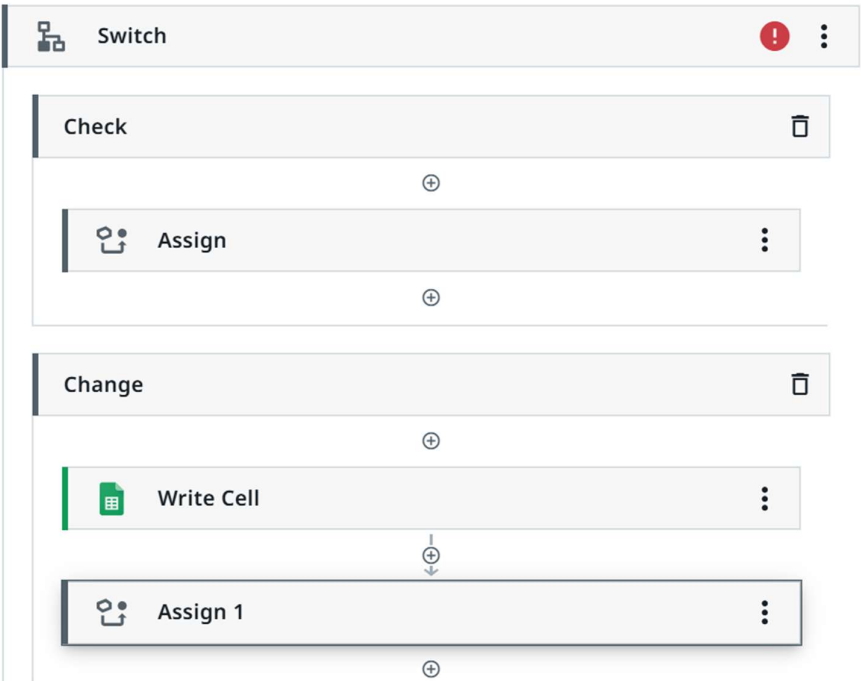
> **In_CustomerDate**

> In_CustomerName

> In_CustomerRequest

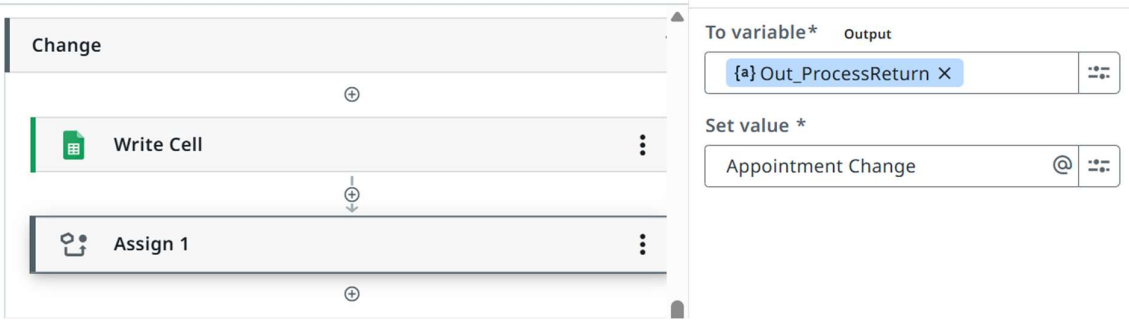
> Out_ProcessReturn

Add activity **Assign**



In the **To variable** field, click **See More** → **Change variable** and select **Out_ProcessReturn**

In the **Set value** field, type in **Appointment Change**



Add a third child, name it **Cancel**. Add activity **Delete Rows (Google Workspace)**
Fill in **Spreadsheet** field as **appointments** and **Range** field as **appointments (Sheet)** accordingly

Cancel

Delete Rows

Google Sheets connection

Spreadsheet * [Use spreadsheet template](#)

appointments

Range *

Sheet1 (Sheet)

Rows *

E.g. 0, 2, 5-7

Delete behavior

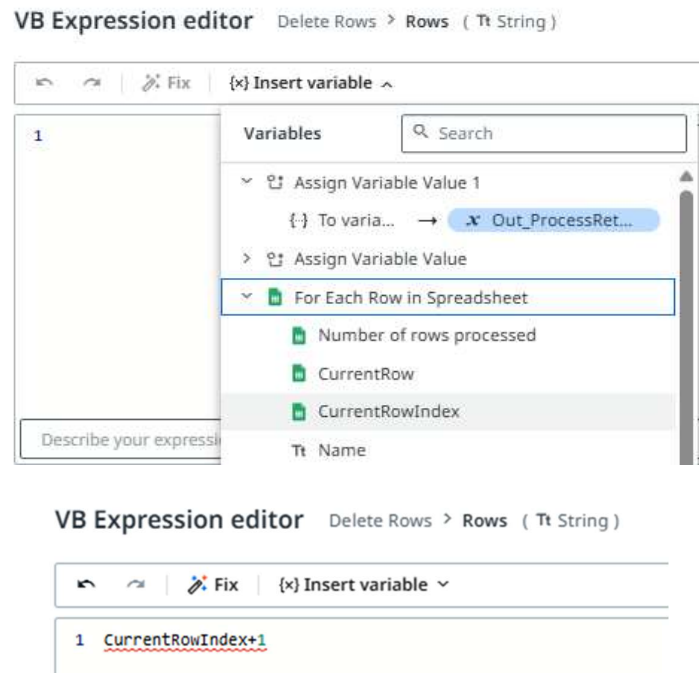
Clear

Output

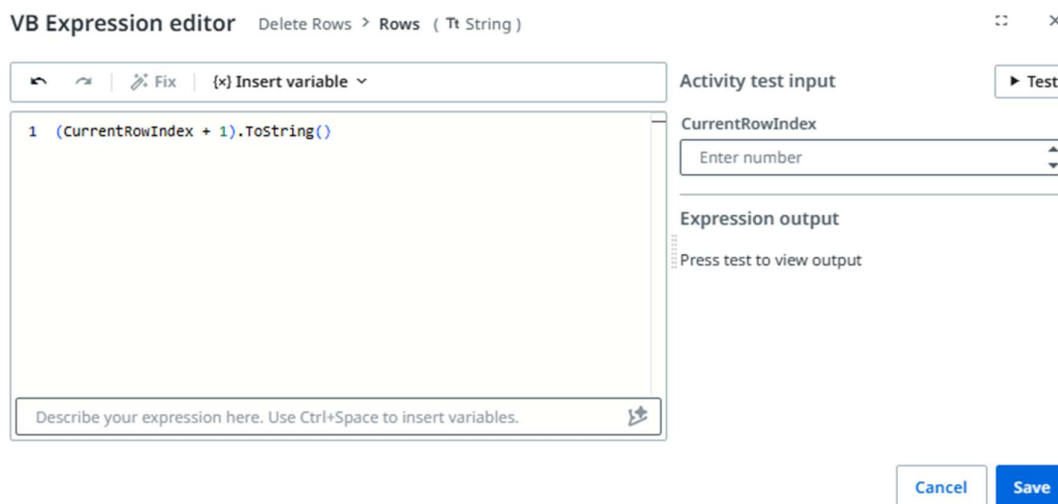
Range information Output

Range information

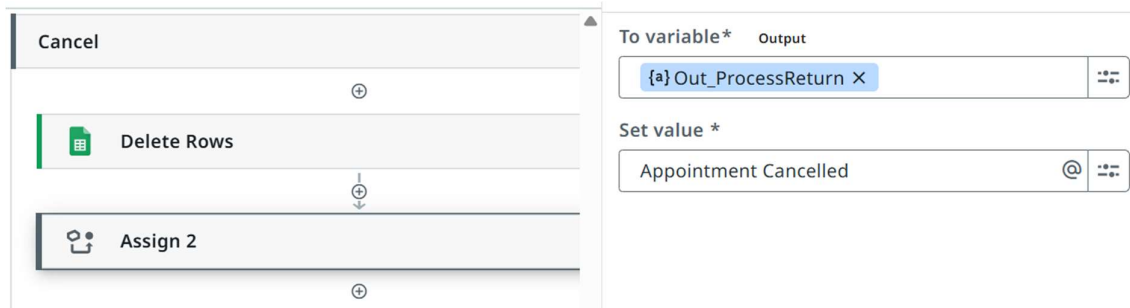
For **Rows** field, click **See more** → **expression editor**. Click **Insert variable** and select **For each row in Spreadsheet** → **CurrentRowIndex**. And type in +1 after the variable.



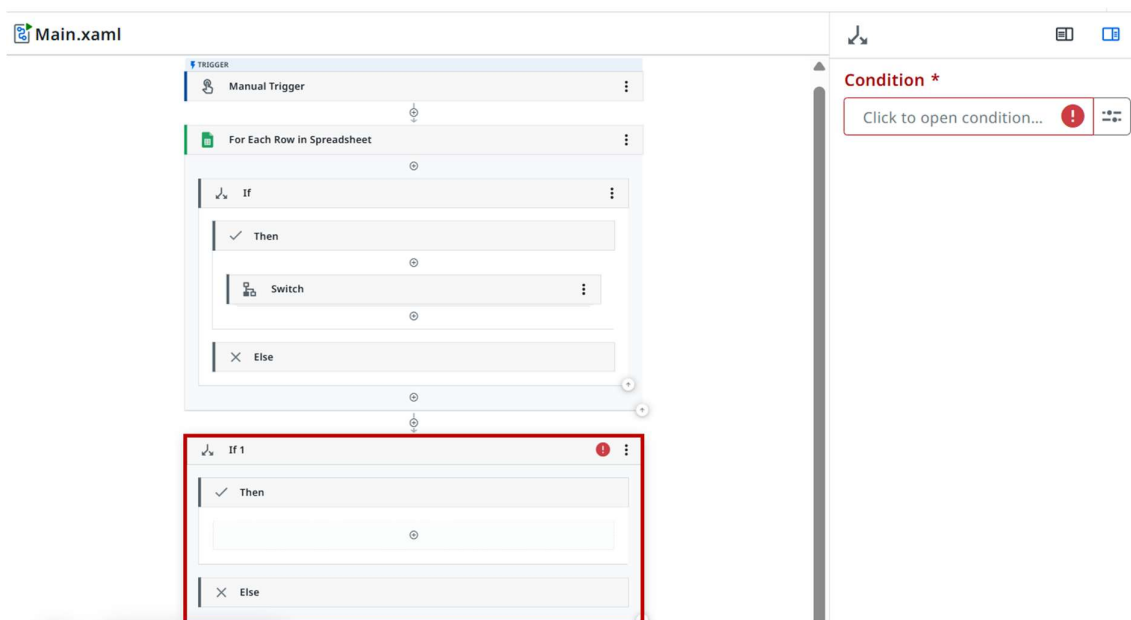
At this point the expression editor should show an error, click the **Fix** button to fix the syntax error. After fixing, you should see the following screen, click **save**.



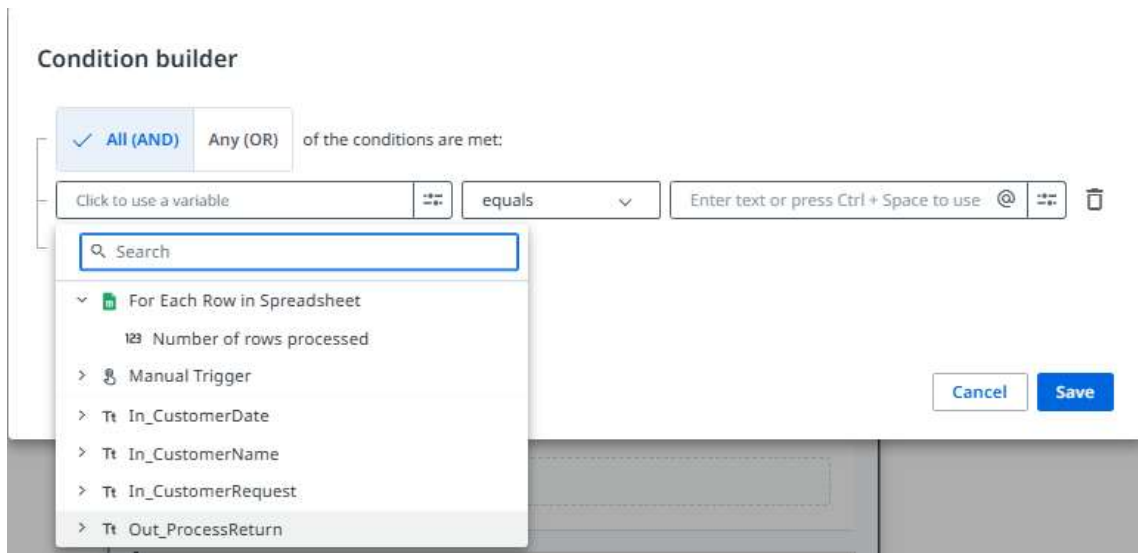
Add **Assign** activity. In **To variable** field, click **See More → Change Variable** and select **Out_ProcessReturn**. In the **Set value** field, type in **Appointment Cancelled**



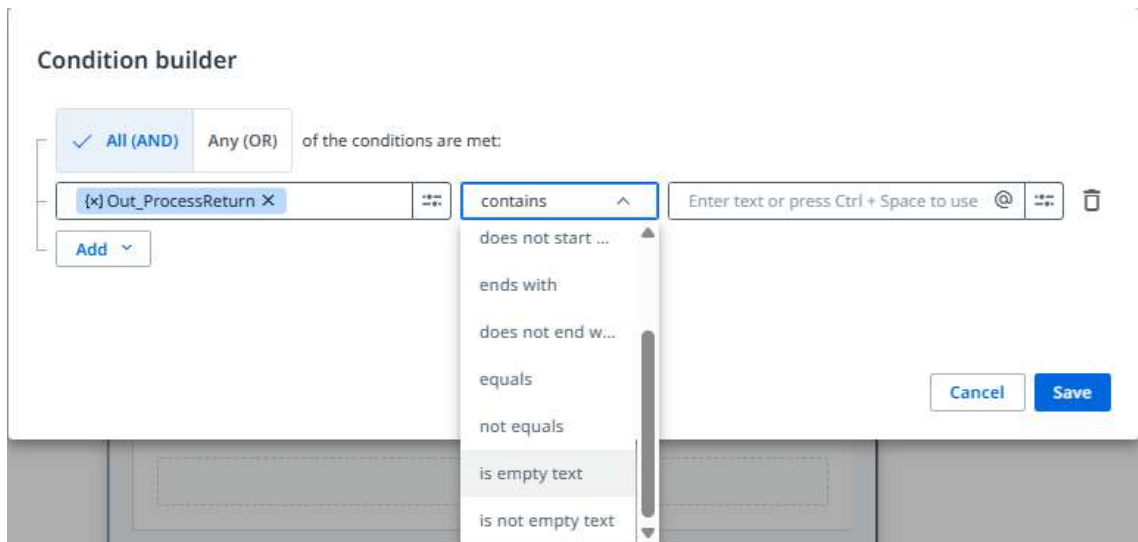
Add a “If” activity (outside For Each Row in Spreadsheet)



Click to open condition builder. Click **See More → Variable** and select **Out_ProcessReturn**



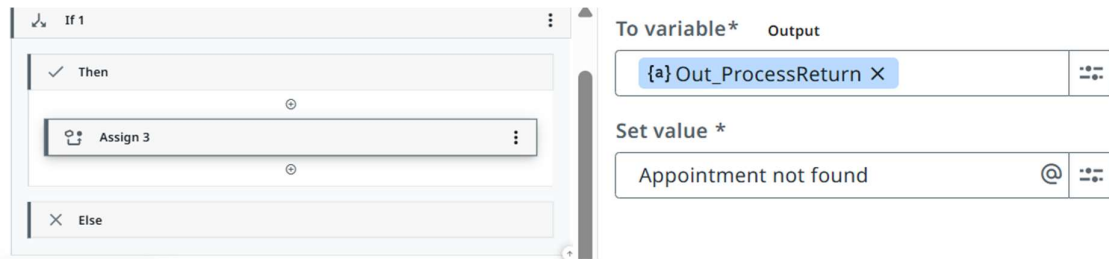
Select **is empty text**



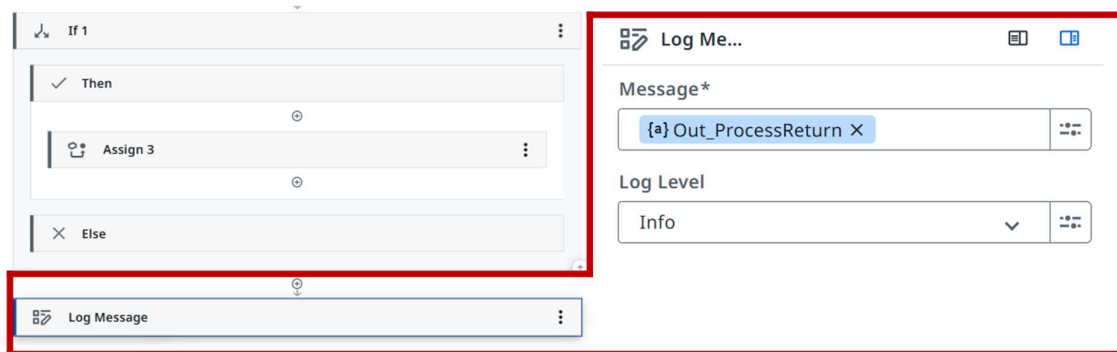
If statement block should look like this, click **Save**



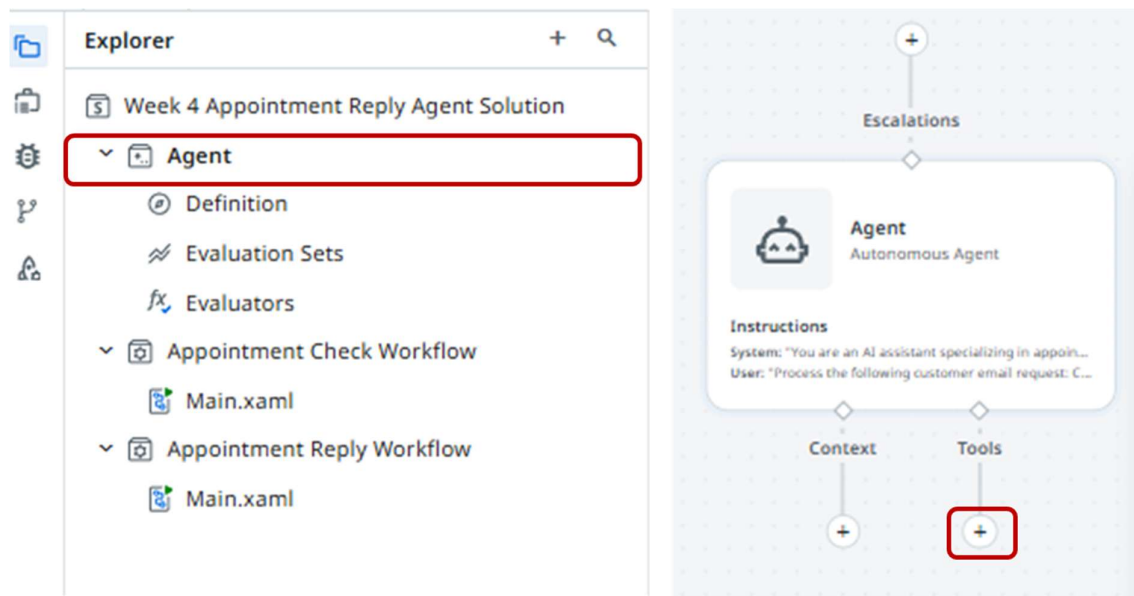
Add **Assign** activity. In the To variable field, click **See more** → **Change variable**, select **Out_ProcessReturn**. In the **Set value** field, type Appointment not found



Add a **log message**. In the Message field, click **See More** → **Variable** and select **Out_ProcessReturn**. In the **Log** level field, select Info from the dropdown menu.



5. Go to **Agent** (Left). Click on **Agent** (Right), click on + Tools



Click on **New tool**. Choose **RPA workflow**

The screenshot shows the 'New tool' dialog on the left and the 'Toolbox' on the right. The dialog has a header 'Agent Autonomous Agent' and 'Instructions' for a system and user. Below are 'Context' and 'Tools' sections, each with a plus icon. A dashed blue line connects the 'Tools' plus icon to a 'New tool' button at the bottom right, which is circled in red. The 'Toolbox' on the right lists various tool types, with 'RPA workflow' highlighted by a red box.

Agent
Autonomous Agent

Instructions
System: "You are an AI assistant specializing in appoin...
User: "Process the following customer email request: C..."

Context + Tools +

New tool

Toolbox

- RPA workflow**
Automated process that connect multiple steps
- API workflow**
Lightweight workflow exposed as API endpoints
- Agent**
Another agent performing a smaller job
- Activity**
Single activity, such as sending an email
- MCP server** Preview
Exchange of context between agents and external tools
- IXP** Preview
IXP models for structured, schema driven data extraction from documents
- Built-in Tools**
Ready-to-use tools with built-in functionality

Select **Appointment Check Workflow**

The screenshot shows the 'Choose tool' dialog on the left and the 'Appointment Check Workflow' tool on the right. The dialog has a header 'Choose tool' and a search bar. Below the search bar, 'Appointment Check Workflow' is highlighted by a red box. The tool on the right is a diagram of an 'Agent Autonomous Agent' with 'Instructions' and 'Context'/'Tools' sections. A dashed blue line connects the 'Tools' plus icon to the 'Appointment Check Workflow' tool, which is circled in red.

Choose tool

← RPA workflow Create new

Search RPA workflows

In current solution (2)

- Appointment Check Workflow**
- Appointment Reply Workflow

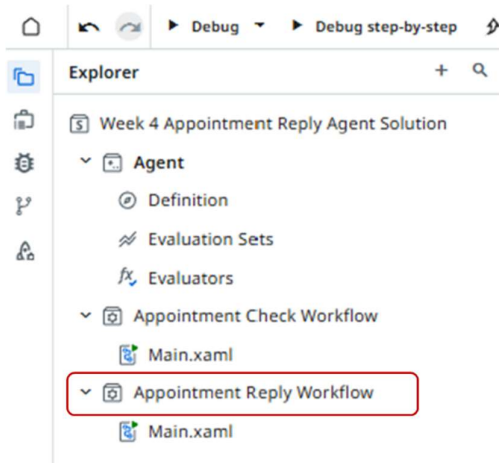
Agent
Autonomous Agent

Instructions
System: "You are an AI assistant specializing in appoin...
User: "Process the following customer email request: C..."

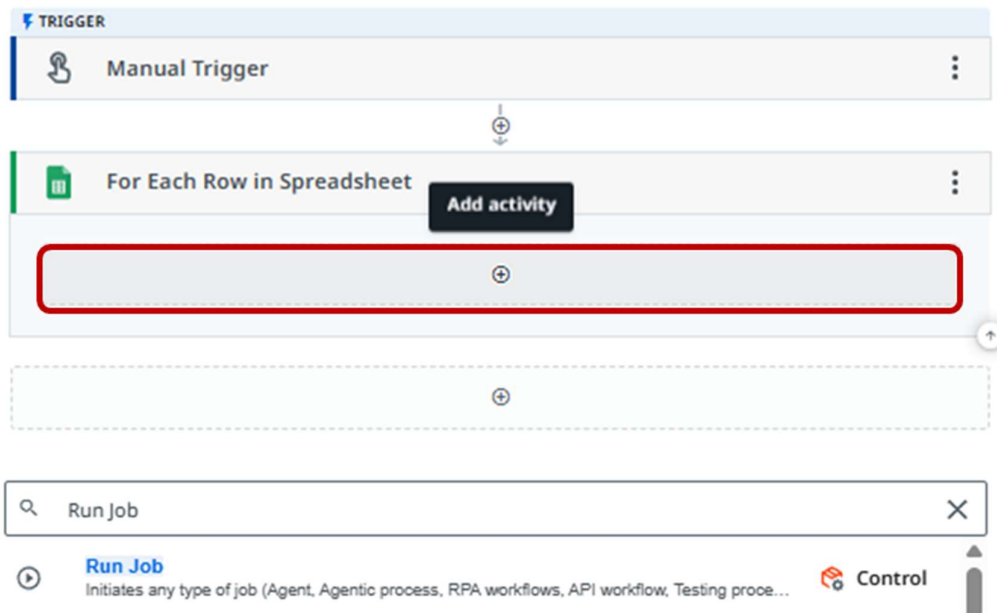
Context + Tools +

Appointment Check Workflow

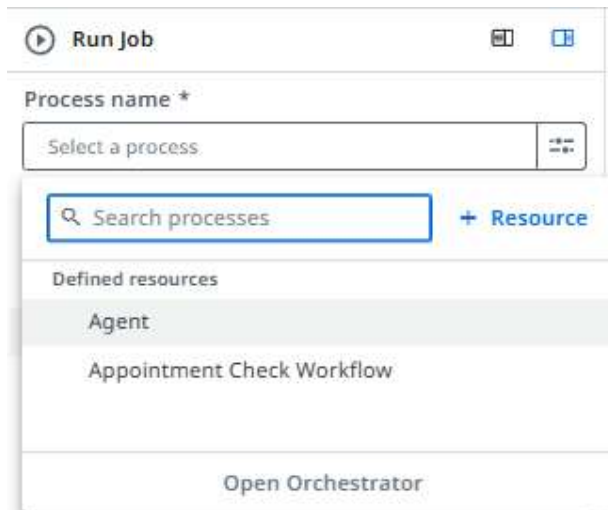
6. Go to **Appointment Reply** workflow



Add **Run job** Activity inside **For each row in Spreadsheet**



In the **Process name** field, select **Agent** from dropdown



Run Job

Process name *

Select a process

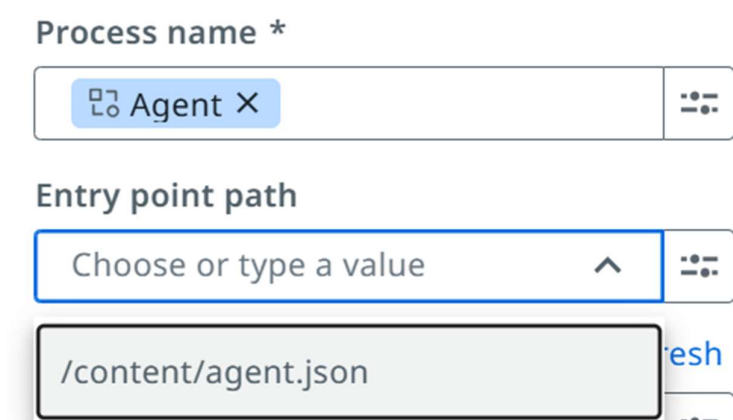
Search processes + Resource

Defined resources

- Agent
- Appointment Check Workflow

Open Orchestrator

Choose the **first entry** in Entry point path.



Process name *

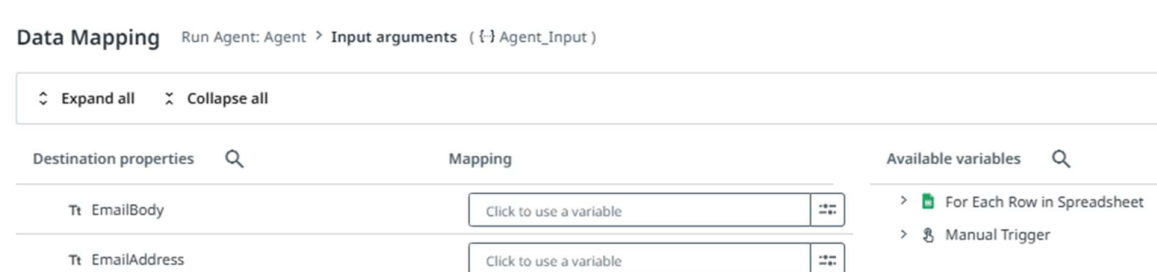
Agent X

Entry point path

Choose or type a value

/content/agent.json

Click on the **Input arguments** field, you should see this screen



Data Mapping Run Agent: Agent > Input arguments ({} Agent_Input)

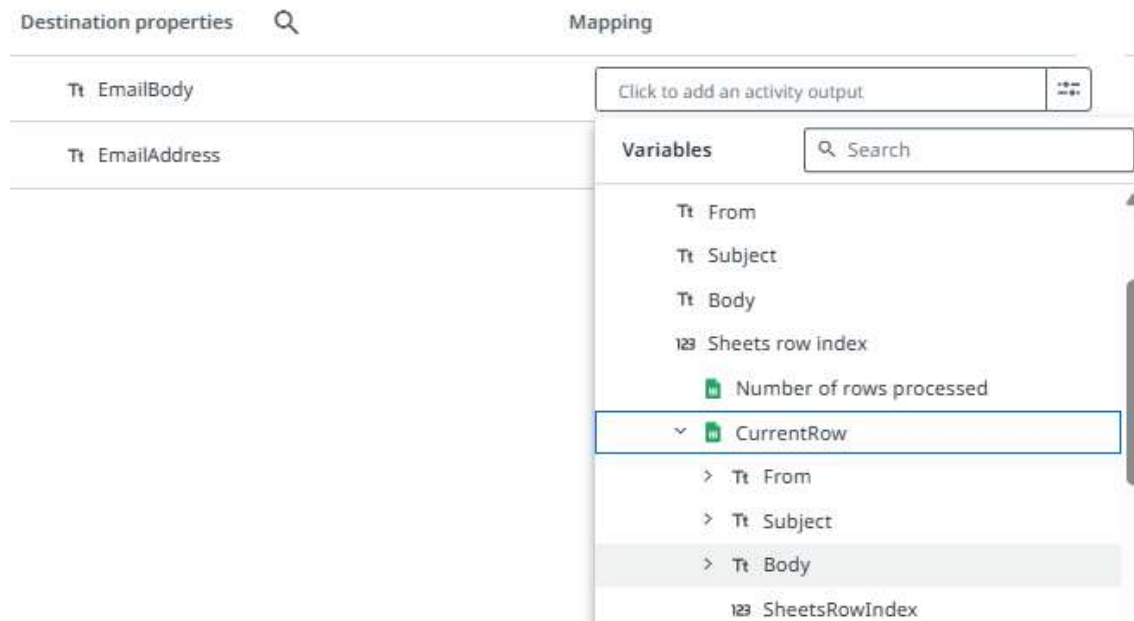
Expand all Collapse all

Destination properties	Mapping
Tt EmailBody	Click to use a variable
Tt EmailAddress	Click to use a variable

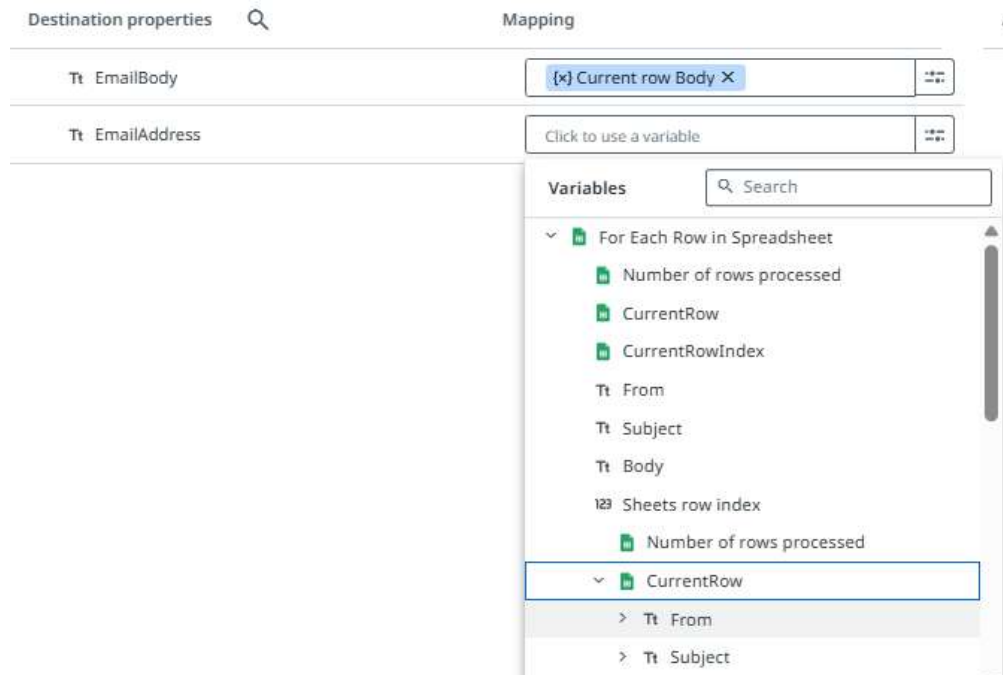
Available variables

- > For Each Row in Spreadsheet
- > Manual Trigger

For EmailBody field, click on **Click to use a variable** → **Show more** → **CurrentRow** → **Body**



For EmailAddress field, click on **Click to use a variable** → **Show more** → **CurrentRow** → **From**

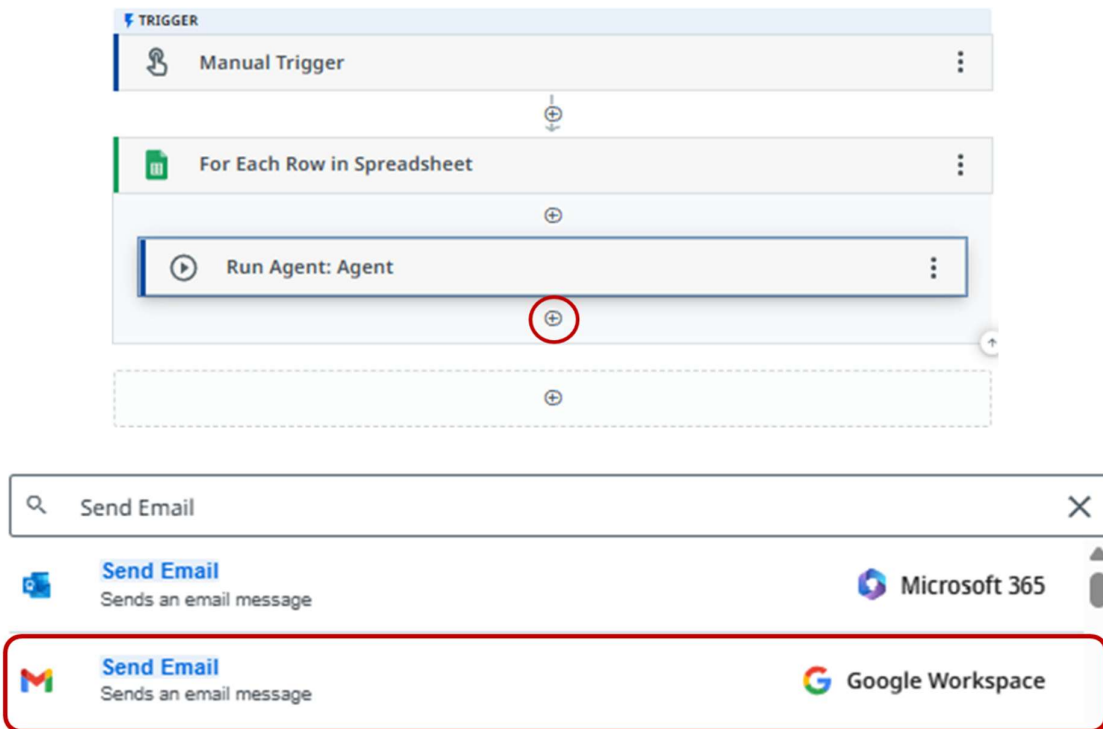


Click **Save**

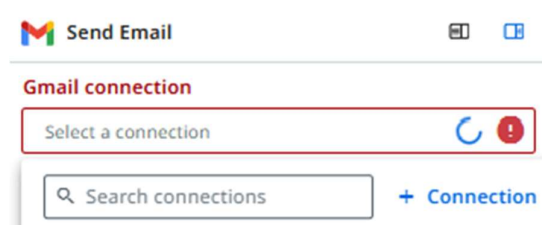
Click “X” to clear Entry point path value. This is a workaround for a minor glitch in the latest release of the platform. If this step is skipped, Agent might run into error.



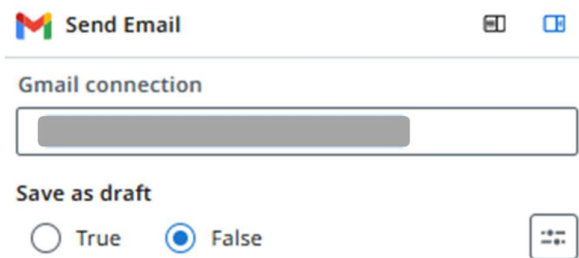
Add **Send Email** (Google Workspace) Activity



Click **Select a connect** and **+ Connection**. Connect your Google account.

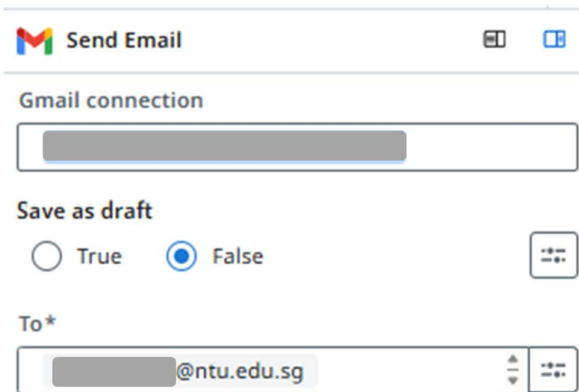


Choose **False** for **Save as Draft** option



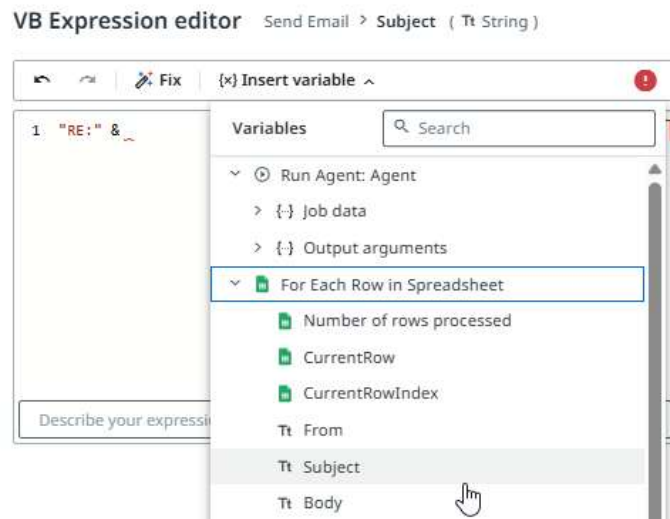
The 'Send Email' configuration window shows the 'Gmail connection' field with a greyed-out dropdown. Below it, the 'Save as draft' section has two radio buttons: 'True' (unselected) and 'False' (selected). A 'See more' icon is visible to the right.

In the **To** field, enter **Your own email address**

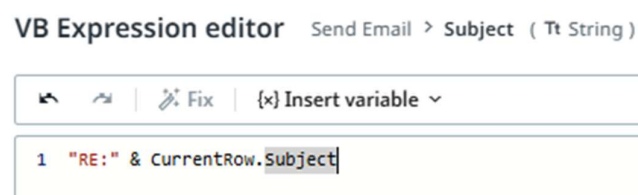


The 'Send Email' configuration window is shown again. The 'To*' field now contains the email address '[redacted]@ntu.edu.sg'. The 'Save as draft' section remains the same.

In the Subject field, click See More → Expression editor. In the editor, Type in **“RE:” & insert variable → For Each Row in Spreadsheet → Subject**



The 'VB Expression editor' window is open for the 'Subject' field. The expression editor on the left contains the text '"RE:" &'. The 'Variables' list on the right is expanded, showing the 'For Each Row in Spreadsheet' category. Under this category, the 'Subject' variable is highlighted by the mouse.



The 'VB Expression editor' window shows the final expression in the editor: '"RE:" & CurrentRow.Subject'. The 'Variables' list is no longer expanded.

In the **Body** field, click **See more** → **Variable** → **Output arguments** → **EmailReply**

The screenshot shows the 'Send Email' configuration window. The 'Body' field is active, and a 'Variables' dropdown menu is open below it. The menu shows a tree structure of variables, with 'EmailReply' selected under 'Output arguments'.

Send Email

Gmail connection

Save as draft

☐ True ☒ False

To*

Subject

f_x "RE:" & CurrentRow.Subject X

Body

Enter text

Variables Search

- Run Agent: Agent
 - Job data
 - Output arguments
 - content
 - EmailReply**
 - For Each Row in Spreadsheet
 - Manual Trigger

7. Double click on “Appointment Reply Workflow” and click “Debug on Cloud”